Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

PENDAHULUAN
TINJAUAN PUSTAKA
[english]babel graphicx framed [normalem]ulem indentfirst amsmath,amsthm,amssymb,amsfonts [italicdiff]physics [T1]fontenc wrapfig lmodern,mathrsfs [inline,shortlabels]enumitem topsep=2pt,itemsep=2pt,parsep=0p [dvipsnames]xcolor [utf8]inputenc [a4paper, top=0.5in,bottom=0.2in, left=0.5in, right=0.5in, footskip=0.3in, includefoot]geometry [most]tcolorbox tikz,tikz-3dplot,tikz-cd,tkz-tab,tkz-euclide,pgf,pgfplots compat=newest multicol [bottom,multiple]footmisc [backend=bibtex,style=numeric]biblatex bibliography hyperref [nameinlink]cleveref
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A Fun Template

Senan Sekhon October 4, 2020

contcol1HTML72E094 contcol2HTML24E2D6 convcol1HTMLC0392B convcol2HTML8E44AD

[title=Contents, fonttitle=,interior style=left color=contcol1!40!white,right color=contcol2!40!w
style=left color=contcol1!80!white,right color=contcol2!80!white,coltitle=black,top=2m
fuzzy shadow,enhanced,breakable] Sample Chapter 1 3
1.1 Some Definitions 3

A Bonus Material 4

[title=Conventions, fonttitle=,interior style=left color=convcol1!40!white,right color=convcol2!40!white,frame style=left color=convcol1!80!white,right color=convcol2!80!whit
fuzzy shadow,enhanced,breakable] F denotes either R or C.
N denotes the set $\{1, 2, 3, ...\}$ of natural numbers (excluding 0).

1 Sample Chapter

Let's dive right in!

1.1 Some Definitions

Definition 1.1 *The **derivative** of a function $f : I \to R$ at $a \in I$ is given by:* $f'(x)=\lim_{x\to a} \frac{f(x)-f(a)}{x-a}$

You know those awesome commutative diagrams?
A [r,"p"] [d,red,"q"'] B [d,"r" red]
C [r,red,"s"' blue] D
The derivative has *nothing* to do with them!

Proposition 1.2 *If f is differentiable at a, then f is continuous at a.*

Bukti. Exercise (but only because this is a template).
The converse of diffcont is not true in general.

1. $f(x) = x$
2. $f(x) = \{ \sin(x) x \geq 0$
   $0x < 0$

Theorem 1.3 *The following statements are true:*

1. *First statement*
2. *Second statement*

Bukti.

1. Trivial.
2. Trivial.

Corollary 1.4 *We are both very lucky to have each other as a collaborator.*

Bukti. We simply note that: $1 \frac{}{1+\frac{1}{1} \gg \frac{1}{1}}$
This corollary is also obvious from empirical evidence.

Lemma 1.5 $(a+b)^2 = a^2 + 2ab + b^2$

Bukti. Expand the left side.

1. It's also kind of obvious.
2. No extra points for guessing what $(a-b)^2$ is.

$(2+4)^2 = 2^2 + 2 \cdot 2 \cdot 4 + 4^2 = 36$

Theorem 1.6 (Pythagoras' Theorem) *If c is the hypotenuse of a right triangle and a and b are the other two sides, then $a^2 + b^2 = c^2$.*

Bukti. Draw a picture and convince yourself.
Pythagoras' theorem helps motivate the study of metric spaces, which you can learn about in [**?**].

A lot of nice integrals can be computed using the residue theorem, see [**?**, Section 5.2].

A Bonus Material

The `talign` and `talign*` environments work like the `align` and `align*` environments, except they render equations in inline size. For example, `\begin{align*}...\end{align*}` yields: $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ While `\begin{talign*}...\end{talign*}` yields: $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ As usual, the purpose of `*` is to prevent numbering of the equation.

Some commands, like `\sumn`, can be used with or without a starting value (the default starting value is 1). For example, `$\sumn\frac{1}{n^2}$` yields $\sum_{n=1}^{\infty} \frac{1}{n^2}$, while `$\sumn[69]\frac{1}{n^2}$` yields $\sum_{n=69}^{\infty} \frac{1}{n^2}$. This can be used in inline mode as well as display mode.